

Systems of Linear Equations (AA HL)

ANSWER KEY



Solving a 2x2 system by elimination

- 1 Solve the system $x + 2y = 8$, $3x - y = 3$ by elimination.
From the second equation, $y = 3x - 3$. Substituting into the first:
 $x + 2(3x - 3) = 8 \Rightarrow x + 6x - 6 = 8 \Rightarrow 7x = 14 \Rightarrow x = 2$. Then $y = 3(2) - 3 = 3$. Solution: $(x, y) = (2, 3)$.
 - 2 Solve the system $3x + 2y = 16$, $x - y = 2$ by elimination.
From the second equation, $y = x - 2$. Substituting into the first:
 $3x + 2(x - 2) = 16 \Rightarrow 3x + 2x - 4 = 16 \Rightarrow 5x = 20 \Rightarrow x = 4$. Then $y = 4 - 2 = 2$. Solution:
 $(x, y) = (4, 2)$.
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Solving a 2x2 system by substitution

- 3 Solve the system $x - 3y = -5$, $2x + y = 4$ by substitution.
From the first equation, $x = 3y - 5$. Substituting into the second:
 $2(3y - 5) + y = 4 \Rightarrow 6y - 10 + y = 4 \Rightarrow 7y = 14 \Rightarrow y = 2$. Then $x = 3(2) - 5 = 1$. Solution:
 $(x, y) = (1, 2)$.
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A system with no solution

- 4 Show that the system $x + y = 4$, $2x + 2y = 9$ has no solution.
Dividing the second equation by 2 gives $x + y = 4.5$, which contradicts the first equation ($x + y = 4$).
Since $x + y$ cannot equal both 4 and 4.5, the system has no solution — the two lines are parallel and distinct.
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A system with infinitely many solutions

- 5 Show that the system $2x - 4y = 6$, $x - 2y = 3$ has infinitely many solutions, and write the general solution using a parameter t .
The first equation is exactly twice the second ($2(x - 2y) = 2(3) \Rightarrow 2x - 4y = 6$), so the two equations describe the same line and carry no independent information. Letting $y = t$, the second equation gives $x = 2t + 3$. General solution: $(x, y) = (2t + 3, t)$ for any $t \in \mathbb{R}$.
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Solving a 3x3 system by elimination

- 6 Solve the system $x + y + z = 6$, $x - y + 2z = 5$, $2x + y - z = 1$.

Adding equation 1 and equation 2 eliminates y : $2x + 3z = 11$ (A). Adding equation 2 and equation 3 eliminates y again: $3x + z = 6$, so $z = 6 - 3x$ (C). Substituting (C) into (A): $2x + 3(6 - 3x) = 11 \Rightarrow 2x + 18 - 9x = 11 \Rightarrow -7x = -7 \Rightarrow x = 1$. Then $z = 6 - 3(1) = 3$. From equation 1: $y = 6 - x - z = 6 - 1 - 3 = 2$. Solution: $(x, y, z) = (1, 2, 3)$.

A 3x3 system with no solution

- 7 Explain why the system $x + y + z = 4$, $2x + 2y + 2z = 10$, $x - y + z = 2$ has no solution.

Dividing the second equation by 2 gives $x + y + z = 5$, which directly contradicts the first equation ($x + y + z = 4$). The two planes $x + y + z = 4$ and $x + y + z = 5$ are parallel and distinct, so no point can satisfy both — the system has no solution regardless of the third equation.

A 3x3 system with infinitely many solutions

- 8 Show that the system $x + y + z = 6$, $x - y + z = 2$, $2x + 2z = 8$ has infinitely many solutions, and write the general solution using a parameter t .

Adding equation 1 and equation 2 gives $2x + 2z = 8$, which is exactly equation 3 — so the third equation is redundant and the system reduces to two independent equations in three unknowns. From equation 1 and equation 2: adding gives $2x + 2z = 8 \Rightarrow x + z = 4$, and subtracting gives $2y = 4 \Rightarrow y = 2$. Letting $z = t$, then $x = 4 - t$. General solution: $(x, y, z) = (4 - t, 2, t)$ for any $t \in \mathbb{R}$.

Classifying a system without fully solving it

- 9 Without solving, determine whether $x + 2y = 5$, $2x + 4y = 9$ has a unique solution, no solution, or infinitely many solutions. Justify your answer using the coefficients.

The coefficients of the second equation are exactly twice those of the first ($2 = 2(1)$, $4 = 2(2)$), but the constant term is not ($9 \neq 2(5) = 10$). The two lines are parallel but distinct, so the system has no solution.

- 10 Without solving, determine whether $x - y = 3$, $2x - 2y = 6$ has a unique solution, no solution, or infinitely many solutions. Justify your answer using the coefficients.

Every coefficient and the constant term of the second equation are exactly twice those of the first ($2 = 2(1)$, $-2 = 2(-1)$, $6 = 2(3)$), so the two equations describe the same line. The system has infinitely many solutions.

Geometric interpretation: where a line of intersection meets a third plane

- 11 The planes $x + y + z = 4$ and $x - y + z = 2$ intersect in the line $(3 - t, 1, t)$. Find the point where this line meets the plane $2x + z = 9$.
- Substitute the line into the plane's equation: $2(3 - t) + t = 9 \Rightarrow 6 - 2t + t = 9 \Rightarrow 6 - t = 9 \Rightarrow t = -3$. The point is $(3 - (-3), 1, -3) = (6, 1, -3)$. Check: $6 + 1 - 3 = 4 \checkmark$, $6 - 1 - 3 = 2 \checkmark$, $2(6) + (-3) = 9 \checkmark$.
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Application: fitting a quadratic through three points

- 12 Find a, b, c such that the parabola $y = ax^2 + bx + c$ passes through $(1, 6)$, $(2, 11)$, and $(3, 18)$, by setting up and solving a system of three linear equations.
- Substituting each point gives $a + b + c = 6$, $4a + 2b + c = 11$, $9a + 3b + c = 18$. Subtracting the first from the second: $3a + b = 5$ (A). Subtracting the second from the third: $5a + b = 7$ (B). Subtracting (A) from (B): $2a = 2 \Rightarrow a = 1$. Then $b = 5 - 3(1) = 2$, and $c = 6 - 1 - 2 = 3$. So $y = x^2 + 2x + 3$ (check: at $x = 3$, $9 + 6 + 3 = 18 \checkmark$).
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Application: a cost word problem

- 13 3 adult tickets and 2 child tickets cost 58 dollars in total. 2 adult tickets and 5 child tickets cost 68 dollars in total. Find the price of an adult ticket and the price of a child ticket by setting up and solving a system of two linear equations.
- Let a and c be the adult and child ticket prices, in dollars. Then $3a + 2c = 58$ and $2a + 5c = 68$. Multiplying the first by 5 and the second by 2: $15a + 10c = 290$ and $4a + 10c = 136$. Subtracting: $11a = 154 \Rightarrow a = 14$. Substituting into the first equation: $3(14) + 2c = 58 \Rightarrow 42 + 2c = 58 \Rightarrow c = 8$. Adult tickets cost 14 dollars; child tickets cost 8 dollars.
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Application: a three-numbers word problem

- 14 The sum of three numbers is 18. The second number is twice the first, and the third number is three times the first. Find the three numbers by setting up and solving a system of three linear equations.
- Let the first number be x . Then the second is $y = 2x$ and the third is $z = 3x$, and $x + y + z = 18$. Substituting: $x + 2x + 3x = 18 \Rightarrow 6x = 18 \Rightarrow x = 3$. So $y = 6$ and $z = 9$ (check: $3 + 6 + 9 = 18 \checkmark$).
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Solving a system with decimal coefficients

- 15 Solve the system $0.4x + 0.3y = 2.5$, $0.2x - 0.1y = 0.5$.
- From the second equation, $y = 2x - 5$. Substituting into the first: $0.4x + 0.3(2x - 5) = 2.5 \Rightarrow 0.4x + 0.6x - 1.5 = 2.5 \Rightarrow x = 4$. Then $y = 2(4) - 5 = 3$. Solution: $(x, y) = (4, 3)$.
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Solving a 2x2 system using Cramer's Rule

- 16 Use Cramer's Rule to solve $2x + 3y = 8$, $x - y = -1$.

The main determinant is $D = 2(-1) - 3(1) = -5$. Replacing the x -column with the constants: $D_x = 8(-1) - 3(-1) = -5$, so $x = D_x/D = 1$. Replacing the y -column: $D_y = 2(-1) - 8(1) = -10$, so $y = D_y/D = 2$. Solution: $(x, y) = (1, 2)$ (check: $2(1) + 3(2) = 8$ ✓, $1 - 2 = -1$ ✓).

A harder 3x3 system

- 17 Solve $x + 2y - z = 3$, $2x - y + z = 7$, $x + y + 2z = 8$.

Adding equation 1 and equation 2 eliminates z : $3x + y = 10$ (A). Doubling equation 2 and subtracting equation 3 also eliminates z : $2(2x - y + z) - (x + y + 2z) = 14 - 8 \Rightarrow 3x - 3y = 6 \Rightarrow x - y = 2$ (B). From (A), $y = 10 - 3x$; substituting into (B): $x - (10 - 3x) = 2 \Rightarrow 4x = 12 \Rightarrow x = 3$. Then $y = 10 - 9 = 1$, and from equation 1, $z = x + 2y - 3 = 3 + 2 - 3 = 2$. Solution: $(x, y, z) = (3, 1, 2)$ (check equation 2: $2(3) - 1 + 2 = 7$ ✓; equation 3: $3 + 1 + 4 = 8$ ✓).

Finding a parameter for infinitely many solutions

- 18 Find the value of k for which the system $3x + ky = 7$, $6x + 4y = 14$ has infinitely many solutions.

The second equation's coefficients and constant must all be exactly twice the first's: $6 = 2(3)$ and $14 = 2(7)$ already hold, so matching the y -coefficients requires $4 = 2k \Rightarrow k = 2$.

Finding a parameter for no solution

- 19 Find the value of k for which the system $x + 3y = 4$, $2x + ky = 9$ has no solution.

For no solution, the lines must be parallel but distinct: the coefficient ratio must match ($2/1 = 2$, so $k/3 = 2 \Rightarrow k = 6$) while the constant ratio does not ($9/4 = 2.25 \neq 2$, so the equations genuinely conflict once $k = 6$).

Geometric interpretation: two parallel planes

- 20 The planes $x + y + z = 1$, $x + y + z = 4$, and $2x - y + z = 3$ have no point in common. Explain why, without needing the third equation.

The first two planes share the same normal vector $(1, 1, 1)$ but have different constant terms (1 versus 4), so they are parallel and distinct — no point can satisfy $x + y + z = 1$ and $x + y + z = 4$ simultaneously. Since no point lies on both, the third equation cannot rescue the system: it has no solution regardless of what the third plane is.

Application: a traffic-flow word problem

- 21** At a road network, flow conservation at two junctions gives $2x - y = 30$ and $x + y = 90$, where x and y are traffic flows in vehicles per hour. Find x and y .

From the second equation, $y = 90 - x$. Substituting into the first:

$2x - (90 - x) = 30 \Rightarrow 3x - 90 = 30 \Rightarrow x = 40$. Then $y = 90 - 40 = 50$. Solution: $(x, y) = (40, 50)$.

Application: balancing a chemical equation

- 22** Balance the combustion equation $aC_3H_8 + bO_2 \rightarrow cCO_2 + dH_2O$ by setting $a = 1$ and using conservation of carbon, hydrogen, and oxygen atoms to set up and solve a system of linear equations for b, c, d .

Carbon: $3a = c \Rightarrow c = 3$. Hydrogen: $8a = 2d \Rightarrow d = 4$. Oxygen: $2b = 2c + d \Rightarrow 2b = 6 + 4 = 10 \Rightarrow b = 5$.

The balanced equation is $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$.

Application: a three-way investment word problem

- 23** 12000 dollars is invested across three accounts paying 3%, 4%, and 5% annual interest. The amount in the 4% account is twice the amount in the 3% account, and the total annual interest earned is 500 dollars. Find the amount invested in each account.

Let a, b, c be the amounts (in dollars) in the 3%, 4%, and 5% accounts. Then $a + b + c = 12000$, $b = 2a$, and $0.03a + 0.04b + 0.05c = 500$. Substituting $b = 2a$ into the first equation gives $c = 12000 - 3a$.

Substituting both into the interest equation gives $0.03a + 0.08a + 0.05(12000 - 3a) = 500$.

Expanding: $0.03a + 0.08a + 600 - 0.15a = 500$. Combining like terms: $-0.04a + 600 = 500$. So $-0.04a = -100$, giving $a = 2500$. Then $b = 5000$ and $c = 12000 - 7500 = 4500$ (check:

$0.03(2500) + 0.04(5000) + 0.05(4500) = 75 + 200 + 225 = 500$ ✓).

Application: recognizing an inconsistent price system

- 24** A shop sells only pens (price p dollars each) and notebooks (price n dollars each). One customer buys 3 pens and 2 notebooks for 16 dollars. Another buys 6 pens and 4 notebooks for 35 dollars. Explain why these two totals cannot both be correct.

The second purchase (6 pens, 4 notebooks) is exactly double the first (3 pens, 2 notebooks) in quantity, so at consistent prices it should cost exactly double: $2 \times 16 = 32$ dollars, not 35. Since $3p + 2n = 16$ and $6p + 4n = 35$ describe parallel but distinct lines (the coefficients double but the constant doesn't), the system has no solution — one of the two totals must be a data error.